Continuous Prob Distributions

1. Rectangular or Uniform Prob Distribution

Defn: A continuous r.v. X is said to have uniform or rectangular prob distn over the interval (a, b) i.e $(-\infty < a < b < \infty)$, if its pdf is given by :

$$f(x; a, b) = \begin{cases} \frac{1}{b-a}, & \text{if } a < x < b \\ 0, & \text{otherwise} \end{cases} \quad \text{---} \quad (*)$$

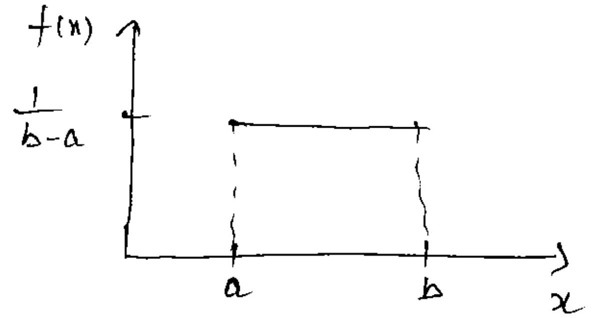
Remark :

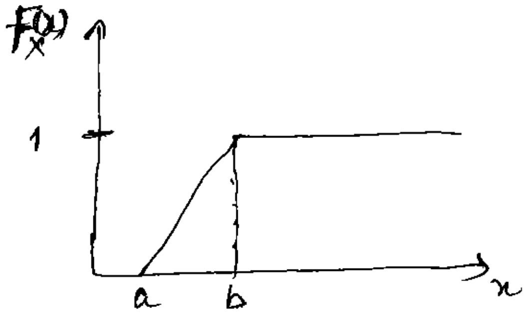
i) $X \sim U(a, b)$

ii) $F_X(x) = \int_a^x f(x)\,dx = \frac{x-a}{b-a}, \quad a < x < b$

iii) $\frac{d}{dx} F_X(x) = f(x) = \frac{1}{b-a} \neq 0$ exists everywhere except the point $x = a$ and $x = b$ and consequently pdf of x is given by $(*)$

iv) Graph of $f(x)$ and $F_X(x)$ are given below





v) $X \sim U(-a, a)$

$$f(x) = \frac{1}{2a}, \quad -a < x < a$$

Moments of Rectangular Distn U(a, b)

$$\mu_r' = E(X^r) = \int_a^b x^r f(x)\,dx = \frac{1}{b-a}\int_a^b x^r dx = \frac{1}{b-a}\left(\frac{b^{r+1} - a^{r+1}}{r+1}\right)$$

Hence mean $\mu_1' = \frac{1}{b-a}\left(\frac{b^2-a^2}{2}\right) = \frac{b+a}{2}$,

$$\mu_2' = \frac{1}{b-a}\left(\frac{b^3-a^3}{3}\right) = \frac{1}{3}(b^2+ab+a^2)$$

$\therefore$ variance $\mu_2 = \mu_2' - \mu_1'^2 = \frac{1}{3}(b^2+ab+a^2) - \left(\frac{1}{2}(b+a)\right)^2$

$$= \frac{1}{12}(b-a)^2$$

MGF of Rectangular (Uniform) dist^n $U(a,b)$

$$M_X(t) = E\,e^{tX} = \int_a^b e^{tx} f(x)\,dx = \int_a^b \frac{e^{tx}}{b-a}\,dx$$

$$= \frac{e^{bt} - e^{at}}{t(b-a)}, \quad t \neq 0$$

Characteristic Function

$$\phi_X(t) = \int_a^b e^{itx} f(x)\,dx = \frac{e^{ibt} - e^{iat}}{it(b-a)}, \quad t \neq 0$$

Ex. If X is uniformly distributed with mean 1 and variance $\frac{4}{3}$, find $P(X<0)$.

if $X \sim U(a,b)$, Mean $= \frac{1}{2}(b+a)$ and

Variance $= \frac{1}{12}(b-a)^2$

Hence $\frac{1}{2}(b+a) = 1$

$\Rightarrow b + a = 2$,

$\frac{1}{12}(b-a)^2 = \frac{4}{3}$

$\Rightarrow (b-a)^2 = 16$

$\Rightarrow (b-a) = \pm 4$

Solving we get $a=-1$ and $b=3$ $(a<b)$

$$\therefore f(x) = \frac{1}{b-a} = \frac{1}{4}, \quad -1<x<3$$

Hence $P(X<0) = \int_{-1}^{0} \frac{1}{4}\,dx = \frac{1}{4}[x]_{-1}^{0} = \frac{1}{4}$.

Ex. Subway trains on a certain line run every half hour between midnight and six in the morning. What is the probability that a man entering the station at a random time during this period will have to wait atleast 20 minutes?

Sol^n. X : denote the waiting time in minutes for the next train.

$$X \sim U(0, 30)$$

$$f(x) = \frac{1}{30}, \quad 0 < x < 30$$

$$P(X \geq 20) = \int_{20}^{30} f(x)\,dx = \int_{20}^{30} \frac{1}{30}\,dx = \frac{1}{30}\Big[x\Big]_{20}^{30}$$

$$= \frac{1}{30}\left[30 - 20\right] = \frac{10}{30} = \frac{1}{3}.$$

GAMMA DISTRIBUTION

Def^n: A r.v. X is said to have a gamma distribution with parameter $\lambda > 0$, if its p.d.f. is given by:

$$f(x) = \begin{cases} \dfrac{e^{-x}\, x^{\lambda-1}}{\Gamma\lambda} ; & \lambda > 0,\ 0 < x < \infty \\ 0, & \text{otherwise} \end{cases}$$

Remark: (i) $X \sim \Gamma\lambda$

ii) $f(x)$ is p.d.f. $\int_0^\infty f(x)\,dx = \frac{1}{\Gamma\lambda}\int_0^\infty e^{-x} x^{\lambda-1}\,dx$

$$= \frac{\Gamma\lambda}{\Gamma\lambda} = 1.$$

iii) gamma dist^n with two parameters.

$$f(x) = \begin{cases} \dfrac{a^\lambda}{\Gamma\lambda}\, e^{-ax}\, x^{\lambda-1} ; & a > 0,\ \lambda > 0,\ 0 < x < \infty \\ 0, & \text{otherwise} \end{cases}$$

$$X \sim \Gamma(a, \lambda), \qquad X \sim \Gamma\lambda = X \sim \Gamma(1, \lambda)$$

iv) $F_X(x) = P(X \leq x) = \int_0^x f(x)\,dx = \frac{1}{\Gamma\lambda}\int_0^x e^{-u} u^{\lambda-1}\,du \cdot x > 0$

MGF of Gamma Distribution $\Gamma\lambda$.

$$M_X(t) = E e^{tX} = \int_0^\infty e^{tx} f(x)\,dx = \frac{1}{\Gamma\lambda}\int_0^\infty e^{tx} e^{-x} x^{\lambda-1}\,dx$$

$$= \frac{1}{\Gamma\lambda}\int_0^\infty e^{-(1-t)x} x^{\lambda-1}\,dx = \frac{1}{\Gamma\lambda}\cdot\frac{\Gamma\lambda}{(1-t)^\lambda}, \quad |t|<1$$

$$M_X(t) = (1-t)^{-\lambda}, \quad |t|<1.$$

Note: For $\Gamma(a,\lambda)$, $\quad M_X(t) = \left(1-\frac{t}{a}\right)^{-\lambda}, \quad t<a$

$$\mu_1' = \text{Mean} = \left.\frac{d M_X(t)}{dt}\right|_{t=0}, \quad \frac{\partial M_X(t)}{\partial t} = (-\lambda)(1-t)^{-\lambda-1}(-1)$$

$$\mu_1' = \left.\frac{\partial M_X(t)}{\partial t}\right|_{t=0} = \lambda.$$

$$\mu_2' = \left.\frac{\partial^2 M_X(t)}{\partial t^2}\right|_{t=0}, \quad \frac{\partial^2 M_X(t)}{\partial t^2} = (-\lambda)(-\lambda-1)(1-t)^{-\lambda-2}(-1)(-1)$$

$$\Rightarrow \mu_2' = \lambda(\lambda+1)$$

$\therefore$ Variance $\mu_2 = \mu_2' - \mu_1'^2 = \lambda(\lambda+1) - \lambda^2 = \lambda$

Hence in $\Gamma\lambda$, Mean = variance = λ.

Additive Property: If $X_i \sim \Gamma\lambda_i$ $(i=1,2\ldots k)$ are independent Γ variates, then $\sum_{1}^{k} X_i \sim \Gamma\sum_1^k \lambda_i$

Proof.
$$M_{X_i}(t) = (1-t)^{-\lambda_i}$$

$$M_{X_1+X_2+\cdots X_k}(t) = \prod_{1}^{k} M_{X_i}(t) = (1-t)^{-\sum_1^k \lambda_i}$$ which is the mgf of gamma variate with parameter $\sum_1^k \lambda_i$.

$$\Rightarrow \sum_{1}^{k} X_i \sim \Gamma\sum_1^k\lambda_i \quad : \text{Holds additive property.}$$

Remark: In general if $X_i \sim \Gamma(a,\lambda_i)$, $i=1,2\ldots k$ are indep. r.vs then $\sum_1^k X_i \sim \Gamma(a, \sum_1^k \lambda_i)$.

Beta Distribution of First Kind

Def^n: A cont. r.v. X is said to follow a beta dist^n of first kind with parameters μ and ν ($\mu>0, \nu>0$), if its p.d.f. is given by:

$$f(x) = \begin{cases} \dfrac{1}{\beta(\mu,\nu)} \cdot x^{\mu-1}(1-x)^{\nu-1} & ; (\mu,\nu)>0, \; 0<x<1 \\ 0, & \text{otherwise.} \end{cases}$$

where $\beta(\mu,\nu) = \dfrac{\Gamma\mu\,\Gamma\nu}{\Gamma\mu+\nu}$, a beta function.

Remark:

I) $X \sim \beta_1(\mu,\nu)$

II) if $X \sim \beta_1(\mu,\nu)$ then $(1-x) \sim \beta_1(\nu,\mu)$

III) $F_X(x) = P(X \le x) = \int_0^x f(x)\,dx.$

IV) if $\mu=1, \nu=1$ $\quad f(x) = \dfrac{1}{\beta(1,1)} = 1, \; 0<x<1$

which is $U(0,1)$.

Constants of $\beta_1(\mu,\nu)$

$$\mu_r' = E(X^r) = \int_0^1 x^r f(x)\,dx = \int_0^1 \frac{1}{\beta(\mu,\nu)} x^{\mu+r-1}(1-x)^{\nu-1}\,dx$$

$$= \frac{1}{\beta(\mu,\nu)} \beta(\mu+r,\nu) = \frac{\Gamma\mu+r\,\Gamma\nu}{\Gamma\mu+r+\nu} \cdot \frac{\Gamma\mu+\nu}{\Gamma\mu\,\Gamma\nu}$$

$$= \frac{\Gamma\mu+r\,\Gamma\mu+\nu}{\Gamma\mu+r+\nu\,\Gamma\mu}$$

In particular $\underset{(\text{Mean})}{\mu_1'} = \dfrac{\mu}{\mu+\nu}$, $\mu_2' = \dfrac{\mu(1+\mu)}{(\mu+\nu)(\mu+\nu+1)}$

Variance $(\mu_2) = \mu_2' - \mu_1'^2 = \dfrac{\mu\nu}{(\mu+\nu)^2(\mu+\nu+1)}$.

Beta distribution of Second kind :

Defn: A cont r.v. X is said to have a beta distribution of the second kind with parameters μ and ν, ($\mu>0, \nu>0$), if its p.d.f. is given by:

$$f(x) = \begin{cases} \frac{1}{B(\mu,\nu)} \cdot \frac{x^{\mu-1}}{(1+x)^{\mu+\nu}} ; & (\mu,\nu)>0,\ 0<x<\infty \\ 0, & \text{otherwise.} \end{cases}$$

Remark : i) $X \sim \beta_2(\mu,\nu)$

ii) Beta distn of second kind is transformed to Beta distn of first kind by the transformation : $1+x = \frac{1}{y} \Rightarrow y = \frac{1}{1+x}$

Thus if $X \sim \beta_2(\mu,\nu)$, then $Y \sim \beta_1(\mu,\nu)$.

Moments

$$\mu_r' = E(X^r) = \int_0^\infty x^r f(x)\,dx = \frac{1}{B(\mu,\nu)} \int_0^\infty \frac{x^{\mu+r-1}}{(1+x)^{\mu+\nu}}\,dx$$

$$= \frac{1}{\beta(\mu,\nu)} \int_0^\infty \frac{x^{(\mu+r)-1}}{(1+x)^{(\mu+r)+\nu-r}}\,dx = \frac{\beta(\mu+r, \nu-r)}{\beta(\mu,\nu)}$$

$$= \frac{\Gamma\mu+r\,\Gamma\nu-r}{\Gamma\mu+\nu} \cdot \frac{\Gamma\mu+\nu}{\Gamma\mu\,\Gamma\nu}$$

$$= \frac{\Gamma\mu+r\,\Gamma\nu-r}{\Gamma\mu\,\Gamma\nu} ; \quad \underline{\nu>r}.$$

In particular $\underset{(\text{Mean})}{\mu_1'} = \frac{\Gamma\mu+1\,\Gamma\nu-1}{\Gamma\mu\,\Gamma\nu} = \frac{\mu\Gamma\mu\,\Gamma\nu-1}{\Gamma\mu\,(\nu-1)\Gamma\nu-1} = \frac{\mu}{\nu-1}, \ \nu>1$

Similarly $\mu_2' = \frac{\mu(\mu+1)}{(\nu-1)(\nu-2)}, \ \nu>2$

$\therefore$ Variance $\underline{\mu_2} = \mu_2' - \mu_1'^2 = \frac{\mu(\mu+\nu-1)}{(\nu-1)^2(\nu-2)}, \ \nu>2.$

Exponential Distribution :

Defn: A Cont. r.v. X is said to follow an exponential distribution with parameter $\theta > 0$, if its p.d.f. is given by :

$$f(x, \theta) = \begin{cases} \theta e^{-\theta x} & ; x \geq 0 \\ 0 , & \text{otherwise} \end{cases}$$

Remark i) $X \sim Exp(\theta)$

ii) $F_X(x) = P(X \leq x) = \int_0^x f(x) dx = \int_0^x \theta e^{-\theta x} dx$

$= 1 - e^{-\theta x} \quad , x \geq 0 .$

MGF of $Exp(\theta)$

$$M_X(t) = E e^{tX} = \int_0^\infty e^{tx} f(x)\, dx = \theta \int_0^\infty e^{tx} e^{-\theta x} dx$$

$$= \theta \int_0^\infty e^{-(\theta - t)x} dx = \frac{\theta}{(\theta - t)} = \left(1 - \frac{t}{\theta}\right)^{-1} , \ \theta > t$$

$$M_X(t) = \left(1 - \frac{t}{\theta}\right)^{-1}$$

$$\mu_1' = \text{Mean} = \frac{\partial M_X(t)}{\partial t}\Big|_{t=0} , \quad \frac{\partial M_X(t)}{\partial t} = (-1)\left(1 - \frac{t}{\theta}\right)^{-2}\left(-\frac{1}{\theta}\right)$$

$$\mu_1' = \frac{\partial M_X(t)}{\partial t}\Big|_{t=0} = \frac{1}{\theta}$$

$$\mu_2' = \frac{\partial M_X(t)}{\partial t^2}\Big|_{t=0} , \quad \frac{\partial^2 M_X(t)}{\partial t^2} = (-1)(-2)\left(1 - \frac{t}{\theta}\right)^{-3}\left(-\frac{1}{\theta}\right)\left(-\frac{1}{\theta}\right)$$

$$\mu_2' = \frac{\partial M_X(t)}{\partial t^2}\Big|_{t=0} = \frac{2}{\theta^2}$$

Hence Variance $\mu_2 = \mu_2' - \mu_1'^2 = \frac{2}{\theta^2} - \frac{1}{\theta^2} = \frac{1}{\theta^2}$

Mean $= \frac{1}{\theta}$, Variance $= \frac{1}{\theta^2}$

Remark. Variance $= \frac{1}{\theta^2} = \frac{1}{\theta} \cdot \frac{1}{\theta} = \frac{\text{Mean}}{\theta}$

$\therefore$ Variance > Mean if $0 < \theta < 1$
Variance = Mean if $\theta = 1$
Variance < Mean if $\theta > 1$

Hence Variance $>, =, \text{or} <$ Mean for different values of the parameter.

Ex ~~Result~~: If $X_1, X_2 \cdots X_n$ are independent r.vs ~~with~~ follow exponential distribution with parameter θ_i; $(i = 1, 2, \cdots n)$ then $Z = \min(X_1, X_2, \cdots X_n)$ has exponential distn with parameter $\sum_{i=1}^{n} \theta_i$.

Proof: $G_Z(z) = P(Z \le z) = 1 - P(Z > z) = 1 - P(\min(X_1, X_2 \cdots X_n) > z)$

$$= 1 - P(X_i > z;\ i = 1, 2 \cdots n) = 1 - \prod_{i=1}^{n} P(X_i > z)$$

($\because X_1, X_2 \cdots X_n$ are independent)

$$= 1 - \prod_{i=1}^{n}\left[1 - P(X_i < z)\right] = 1 - \prod_{i=1}^{n}\left[1 - F_{X_i}(z)\right]$$

$$= 1 - \prod_{i=1}^{n}\left[1 - (1 - e^{-\theta_i z})\right] = 1 - e^{-\left(\sum_{i=1}^{n}\theta_i\right) z}$$

$$\therefore g_Z(z) = \frac{d}{dz} G_Z(z) = \left(\sum_{i=1}^{n}\theta_i\right) \exp\left(-\sum_{i=1}^{n}\theta_i\right) z \quad , z > 0$$

Hence $Z = \min(X_1, X_2 \cdots X_n) \sim Exp\left(\sum_{i=1}^{n}\theta_i\right)$

Remark: if $X_i\ (i = 1, 2, \cdots n)$ are iid with parameter θ then $Z = \min(X_1, X_2 \cdots X_n) \sim Exp(n\theta)$.

Ex. Show that the exponential distribution 'lacks memory' i.e if $X \sim Exp(\theta)$, then for every constant $a \geq 0$, one has $P(Y \le x \mid X \geq a) = P(X \le x)\ \forall\ x$, where $Y = X - a$.

Soln: If $X \sim Exp(\theta)$, $f(x) = \theta e^{-\theta x}$; $\theta > 0, 0 < x < \infty$

We have $P(Y \le x \cap X \geq a) = P(X - a \le x \cap X \geq a)$

$$= P(X \le x + a \cap X \geq a) = P(a \le X \le a + x)$$

$$= \theta \int_a^{a+x} e^{-\theta x} dx = \theta \cdot \frac{(-1)}{\theta}\left[e^{-\theta x}\right]_a^{a+x} = e^{-a\theta}(1 - e^{-\theta x})$$

and $P(X \geq a) = \theta \int_a^{\infty} e^{-\theta x} dx = e^{-a\theta}$

$$\therefore P(Y \le x \mid X \geq a) = \frac{P(Y \le x \cap X \geq a)}{P(X \geq a)} = 1 - e^{-\theta x} \quad — (*)$$

$$P(X \le x) = \theta \int_0^x e^{-\theta x}\, dx = 1 - e^{-\theta x} \quad \text{---} \quad (**)$$

Hence from (*) & (**) we get $P(Y \le x \mid X \geq a) = P(X \le x)$

i.e exponential dist^n lacks memory.

WEIBUL Distribution

A random variable X has a weibul dist^n with three parameters $c>0$, $\alpha>0$ and μ, if the r.v. $Y = \left(\frac{X-\mu}{\alpha}\right)^c$ has the exponential dist^n with p.d.f. $f_Y(y) = e^{-y}$, $y>0$.

Def^n: A cont. r.v. X has a weibul dist^n with parameters $c(>0)$, $\alpha(>0)$ and μ if its p.d.f. is:

$$f(x; c, \alpha, \mu) = \frac{c}{\alpha}\left(\frac{x-\mu}{\alpha}\right)^{c-1} \exp\left\{-\left(\frac{x-\mu}{\alpha}\right)^c\right\}; \quad x>\mu,\ c>0.$$

The standard weibul dist^n ~~which depends~~ is obtained by taking $\alpha = 1$ and $\mu = 0$, so that pdf of weibul dist^n depends only on a single parameter c is given by

$$f_X(x) = c\, x^{c-1} \exp(-x^c) \; ; \quad x>0,\ c>0.$$

Moments: Moments of standard weibul dist^n:

When $\alpha = 1$, $\mu = 0$, $Y = X^c$ which has the exponential dist^n. We have $\mu_r' = E(X^r) = E(Y^{\frac{1}{c}})^r = E\, Y^{\frac{r}{c}}$

$$= \int_0^\infty e^{-y} y^{\frac{r}{c}}\, dy \Rightarrow \mu_r' = \Gamma\left(\frac{r}{c}+1\right)$$

$$\text{Mean } (\mu_1') = E(X) = \Gamma\left(\frac{1}{c}+1\right)$$

$$V(X) = \mu_2 = \mu_2' - \mu_1'^2 = \Gamma\left(\frac{2}{c}+1\right) - \left(\Gamma\left(\frac{1}{c}+1\right)\right)^2$$

CAUCHY DISTRIBUTION

Defn: A r.v. X is said to follow a standard Cauchy distribution if its p.d.f. is given by

$$f_X(x) = \frac{1}{\pi(1+x^2)} \; ; \; -\infty < x < \infty$$

and X is termed as standard Cauchy ~~variate~~ variate.

More generally, Cauchy distribution with parameters λ and μ has the p.d.f.:

$$g_Y(y) = \frac{\lambda}{\pi[\lambda^2 + (y-\mu)^2]} \; ; \; -\infty < y < \infty \; ; \; \lambda > 0$$

If $Y \sim C(\lambda, \mu)$, then $X = \frac{Y-\mu}{\lambda} \sim C(1,0)$

1. Characteristic Function of Standard Cauchy Distribution

$$\phi_X(t) = E e^{itX} = \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{e^{itx}}{1+x^2}\,dx = e^{-|t|}$$

Remarks: If $Y \sim C(\lambda, \mu)$

$$X = \frac{Y-\mu}{\lambda} \sim C(1,0) \Rightarrow Y = \mu + \lambda X$$

$$\therefore \phi_Y(t) = E e^{itY} = E e^{it(\mu + \lambda X)} = e^{it\mu}\phi_X(t\lambda)$$

$$= e^{it\mu} e^{-\lambda|t|} = e^{it\mu - \lambda|t|} \; ; \; \lambda > 0$$

2. Additive Property:

If X_1 and X_2 are independent Cauchy variates with parameters (λ_1, μ_1) and (λ_2, μ_2) respectively then $X_1 + X_2$ is a Cauchy variate with parameter $(\lambda_1+\lambda_2, \mu_1+\mu_2)$.

$$\phi_{X_j}(t) = e^{it\mu_j - \lambda_j|t|} \; ; \; j = 1, 2$$

$$\phi_{X_1+X_2}(t) = \phi_{X_1}(t)\cdot\phi_{X_2}(t) = e^{it(\mu_1+\mu_2) - (\lambda_1+\lambda_2)|t|}$$

And the ~~~~ result follows by uniqueness theorem of characteristic functions.

If $Y \sim C(\lambda, \mu)$

3. $\phi_X(t) = e^{i\mu t - \lambda|t|}$; $\lambda > 0$

$\frac{\partial \phi_X(t)}{\partial t} = \phi'_X(t)$ does not exist at $t = 0$, the mean of the cauchy distribution does not exist.

In general μ_r, $(r \geq 2)$ do not exist.

4. Let $X_1, X_2 \cdots X_n$ be a sample of n independent observations from a standard Cauchy distribution and $\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$, then

$$\phi_{\overline{X}}(t) = \phi_{\Sigma X_i}(t/n) = \prod_{j=1}^{n} \phi_{X_j}(t/n) = (\phi_{X_j}(t/n))^n$$

$$= \left(e^{-|\frac{t}{n}|}\right)^n = e^{-|t|} = \phi_X(t). \quad (\text{Since } X_j'\text{s are iid})$$

Hence by uniqueness theorem of characteristic functions, we have:

" The arithmetic mean $\overline{X}$ of sample $X_1, X_2, \cdots X_n$ of independent observations from a standard cauchy distribution is also a standard cauchy variate. In other words, the arithmetic mean of a random sample of any size yields exactly as much information as a single determination of X "

This implies that the sample mean $\overline{X}$ of random sample of size n, as an estimate of pop^n mean does not improve with increasing n, which contradicts the weak Law of Large Number (WLLN).

Remark: The role of cauchy distribution in statistical theory often lies in providing counter examples. e.g. it is often quoted as a distribution for which moments do not exist. it also provides an example to show that $\phi_{X+Y}(t) = \phi_X(t)\phi_Y(t)$ does not imply that X and Y are independent.

Ex. If X have a standard Cauchy distribution. Find the p.d.f. of X^2 and identify its distribution.

Given $f(x) = \frac{1}{\pi}\cdot\frac{1}{(1+x^2)}$, $-\infty < x < \infty$

The distribution function of $Y = X^2$ say $G_Y(y)$ is

$$G_Y(y) = P(Y \le y) = P(X^2 \le y) = P(-\sqrt{y} \le X \le \sqrt{y})$$

$$= \int_{-\sqrt{y}}^{\sqrt{y}} f(x)\,dx = 2\frac{1}{\pi}\int_0^{\sqrt{y}} \frac{dx}{1+x^2} = \frac{2}{\pi}\tan^{-1}(\sqrt{y}),\ 0 < y < \infty$$

The p.d.f $g_Y(y)$ of Y is given by

$$g_Y(y) = \frac{d}{dy}\left[G_Y(y)\right] = \frac{2}{\pi}\cdot\frac{1}{(1+y)}\cdot\frac{1}{2\sqrt{y}} = \frac{1}{\pi}\frac{y^{-\frac{1}{2}}}{(1+y)}$$

$$= \frac{1}{\beta(\frac{1}{2},\frac{1}{2})}\cdot\frac{y^{\frac{1}{2}-1}}{(1+y)^{\frac{1}{2}+\frac{1}{2}}},\quad y > 0$$

This is the p.d.f. of Beta dist^n of second kind with parameters $(\frac{1}{2},\frac{1}{2})$.

i.e $X^2 \sim \beta_2(\frac{1}{2},\frac{1}{2})$.

Ex. If $X \sim N(0,1)$ and $Y \sim N(0,1)$ be independent random variables, then $\frac{X}{Y}$ follows standard Cauchy ~~variate~~ dist^n.

i.e $Z = \frac{X}{Y}$

$$f(z) = \frac{1}{\pi}\cdot\frac{1}{(1+z^2)},\quad -\infty < Z < \infty.$$